Grade

Tecnologico de Monterrey

Partial Exam 2 A

2026-Jan-May

Course: Calculus I **ID-number:** ______________

Professor: Fco. Javier Vazquez Tavares **Date:** 24/03/2026

Last Names and Names: ________________________________ **Group:** 201

"*In accordance with Tec de Monterrey Student Cod of Honor, in this exam I agree to conduct myself guided by academic honestly and not to use unauthorized or illegal means to do so.*"

Signature: ______________

I Multiple option section. Read the following questions and identify the option that best answers each, then write ts letter on the line.

1. (4 **points**) ______ Identify the quotient rule for derivatives.

(a) $\left(\frac{u}{v}\right)' = \frac{v'u - u'v}{u^2}$ (b) $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$ (c) $\left(\frac{u}{v}\right)' = \frac{uv' - u'v}{v^2}$ (d) $\left(\frac{u}{v}\right)' = \frac{u'v - v'u}{u^2}$.

2. (4 **points**) ______ Identify the product rule for derivatives

(a) $(uv)' = u' + v'$ (b) $(uv)' = u'v + uv'$ (c) $(uv)' = uv' + uv'$ (d) $(uv)' = u'v + u'v$

3. (4 **points**) ______ Identify the chain rule for derivatives

(a) $[u(v(x))]' = v'(u(x))$ (b) $[u(v(x))]' = v'(u(x))u'(x)$ (c) $[u(v(x))]' = u'(v(x))v'(x)$ (d) $[u(v(x))]' = u'(v(x))$

4. (4 **points**) ______ Which of the following derivatives is correct

(a) $\frac{\mathrm{d}}{\mathrm{d}x}n^x = n^x \ln(n)$ (b) $\frac{\mathrm{d}}{\mathrm{d}x}n^x = \frac{1}{x}n^x$ (c) $\frac{\mathrm{d}}{\mathrm{d}x}n^x = xn^x$ (d) $\frac{\mathrm{d}}{\mathrm{d}x}n^x = ne^{x-1}$

5. (4 **points**) ______ Which of the following derivatives is correct

(a) $\frac{\mathrm{d}}{\mathrm{d}x}\sin(x) = \cos(x)$ (b) $\frac{\mathrm{d}}{\mathrm{d}x}\sin(x) = -\sin(x)$ (c) $\frac{\mathrm{d}}{\mathrm{d}x}\sin(x) = \sin(x)$ (d) $\frac{\mathrm{d}}{\mathrm{d}x}\sin(x) = -\cos(x)$

6. (4 **points**) ______ Which of the following derivatives is correct

(a) $\frac{\mathrm{d}}{\mathrm{d}x}\cos(x) = \cos(x)$ (b) $\frac{\mathrm{d}}{\mathrm{d}x}\cos(x) = -\sin(x)$ (c) $\frac{\mathrm{d}}{\mathrm{d}x}\cos(x) = \sin(x)$ (d) $\frac{\mathrm{d}}{\mathrm{d}x}\cos(x) = -\cos(x)$

7. (4 **points**) ______ Which of the following derivatives is correct

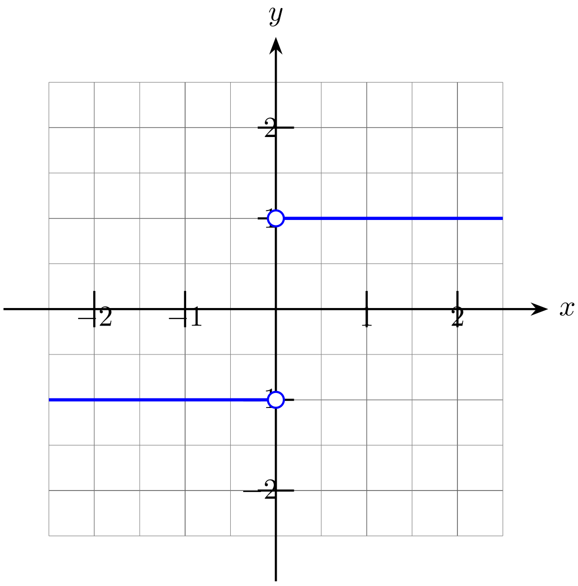
(a) $\frac{\mathrm{d}}{\mathrm{d}x}\log_n(x) = \frac{1}{\ln(x)}$ (b) $\frac{\mathrm{d}}{\mathrm{d}x}\log_n(x) = \frac{1}{x\ln(n)}$ (c) $\frac{\mathrm{d}}{\mathrm{d}x}\log_n(x) = \frac{\ln(x)}{x}$ (d) $\frac{\mathrm{d}}{\mathrm{d}x}\log_n(x) = \frac{1}{x\ln(x)}$

8. (4 **points**) ______ Which of the following options is the implicit derivative of $y^2 = x$

(a) $\frac{\mathrm{d}x}{\mathrm{d}y} = 0$ (b) $\frac{\mathrm{d}x}{\mathrm{d}y} = 2y - 1$ (c) $\frac{\mathrm{d}x}{\mathrm{d}y} = 1$ (d) $\frac{\mathrm{d}x}{\mathrm{d}y} = 1 - 2y$

II Limits. Compute the following limits using the numeric, graphical or algebraic method.

1. (5 **points**) Applying the graphical technique to compute the limit when $x \to 0$. Argument if the limit converges. Use the notion of limit from the left and right.



2. (5 **points**) Compute the limit of the function $f(x) = \frac{\sin(x)}{x}$ as $x \to 0$ using the numeric method.

3. (5 **points**) Compute the limit of the function $f(x) = \frac{6x^2+4}{5x^2-2}$ as $x \to \infty$ using the algebraic method.

4. (5 **points**) Compute the derivative of the function $f(x) = |x|$ at $x = 0$ using the definition of derivatives, $\frac{\mathrm{d}f(x)}{\mathrm{d}x} = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$.

III Derivatives by rules. Use the rules of derivatives that can be usefull to compute the derivatives of the following functions.

1. (8 **points**) Compute the derivative of the function $f(x) = e^{-x^2} \sin^2(x)$.

2. (8 **points**) Compute the derivative of the function $g(x) = \log\left[xe^{-x}\right]$.

IV Critical points. In this section you task is to find <u>all</u> critical points of the functions in the given context.

1. (8 **points**) An open box is made from a 5×5 cm sheet by cutting squares of side x from the corner. The volume of the box is given by $V(x) = x(5-2x)^2$. Find the length of the squares that achieves the greater volume.

2. (8 **points**) Find the critical point of the following function $f(x) = 6x \cdot e^{(3/2)x}$.

V Implicit derivatives.

1. (8 **points**) Compute the implicit derivative of the function $x^3 + y^3 = 6xy$.

IV High order derivatives. Compute the second order derivative of the following functions and reduce them as muach as possible.

1. (8 **points**) $f(x) = e^{x+3}$